

Vancouver Community Health Resource Allocation

Healthcare Planning & Data Analytics Portfolio Case Study | Carlos Nakagomi

Independent portfolio analysis using public/open government data - not affiliated with or commissioned by Vancouver Coastal Health.

Business Objective

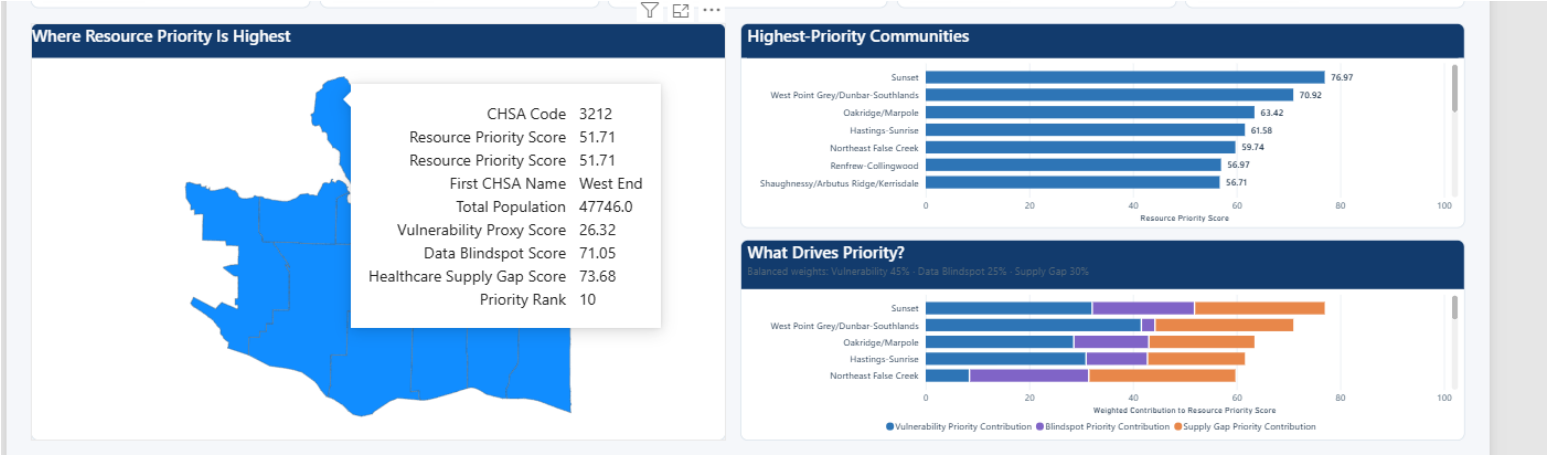
Combine Vancouver community demographics, census non-response, mapped healthcare-facility locations and geographic boundaries to identify comparative planning priorities and test transparent hypothetical resource-allocation scenarios.

663.9K	19	153	2.30	6.51%
Population	Vancouver CHSAs	Mapped facilities	Facilities / 10K	Weighted GNR

Dataset Origin & Analytical Inputs

Source	Dataset / input	Role in the analysis
Statistics Canada	2016 Census / Census Profile	Population, age structure and census non-response indicators for community profiles and the age-structure vulnerability proxy.
Statistics Canada / Government of Canada	Open Database of Healthcare Facilities (ODHF)	Facility names, types and coordinates; spatially mapped to CHSAs as a location-based supply proxy.
Government of British Columbia	Community Health Service Area (CHSA) geography	Planning geography used for spatial aggregation across 19 Vancouver-focused CHSAs.
Statistics Canada	Vancouver Census Subdivision boundary (5915022)	Authoritative city boundary used to validate scope and retained facility locations.

Executive Dashboard View



Power BI view combining community priority geography, ranked communities and weighted vulnerability, census-uncertainty and mapped supply-gap components. The visible tooltip demonstrates community-level drill-down.

Key Findings

Sunset	Highest Balanced Resource Priority Score (76.97); first or second across every tested weighting scenario.
West Point Grey / Dunbar-Southlands	Highest age-structure vulnerability proxy; 0.57 mapped facilities per 10,000 residents.
Downtown Eastside	Highest Long-Form GNR (13.5%); census-response uncertainty, not socioeconomic vulnerability.
Northeast False Creek	Zero mapped ODHF facilities and largest positive illustrative budget shift (+\$160,656.11); not proof of unmet need.

Scope, Data Quality & Validation

Geographic scope 19 CHSAs meeting the Vancouver City overlap rule; UBC CHSA 3243 excluded.	Facility mapping 153 canonical mapped facilities assigned by point-in-polygon; 0 null retained assignments.
Coordinate quality 29 Vancouver candidate facilities lacked coordinates; 1 retained facility used a 25 m edge-tolerance assignment.	Validation 13 automated controls passed, including exact reconciliation to \$10M, 50 clinical FTE and 1,000 community-care units.

Planning Logic, Interpretation & Technical Delivery

The model keeps demographic vulnerability, census uncertainty and mapped facility-location scarcity distinct, then combines them transparently for scenario analysis.

PUBLIC DATA	GEOGRAPHIC VALIDATION	COMMUNITY PROFILE	PRIORITY & ALLOCATION
Census, ODHF and authoritative boundaries.	Vancouver CSD + CHSA overlap + point-in-polygon.	Population, age structure and GNR.	Priority score + hypothetical allocation.

Priority & Allocation Logic

Component	Scenario logic	Interpretation
Resource Priority	45% Vulnerability Proxy + 25% Data Blindspot + 30% Healthcare Supply Gap	Balanced comparative planning score.
Hypothetical Allocation	60% Population Share + 40% Resource Priority Share	Illustrative scenario logic, not a VCH operational rule.

Management Interpretation

Signal	Planning use	Required follow-up
Sunset	Further operational investigation	Validate with current capacity, utilization, wait-time and patient-flow evidence.
West Point Grey / Dunbar-Southlands	Review service-access context	Account for cross-boundary access and actual service capacity.
Northeast False Creek	Validate service mapping	Investigate zero mapped ODHF facilities without inferring unmet need from counts alone.
Downtown Eastside	Supplement census evidence	Use local/administrative evidence because 13.5% GNR indicates higher census uncertainty.

Sensitivity Testing

Scenario	Weights: Vulnerability / Blindspot / Supply Gap	Spearman vs Balanced
Balanced	45% / 25% / 30%	1.000
Need-focused	60% / 10% / 30%	0.869
Uncertainty-aware	35% / 35% / 30%	0.918
Supply-gap-focused	35% / 15% / 50%	0.892

Analytical Guardrails

Vulnerability Proxy Age structure only; not multidimensional deprivation or clinical need.	GNR Census-response uncertainty, not disadvantage.
Facility counts Location proxy, not capacity, access, ownership, staffing, utilization, quality or outcomes.	Allocation Hypothetical portfolio scenario, not actual VCH budgets, staffing or recommendations.

Technical Delivery & Reproducibility

Stage	Delivery
Data understanding & quality	Profile census variables, GNR, facility records, coordinates and authoritative boundaries.
Geospatial processing	Validate Vancouver scope and assign facility points to CHSAs.
Python & SQL	Build profiles, scores, scenario allocations, analytical views and validation controls.
Power BI	Semantic model, Power Query, DAX, standalone PBIX and source-controlled PBIP/PBIR/TMDL.
Validation	13 automated controls; exact \$10M / 50 FTE / 1,000 care-unit reconciliation.

Published Portfolio

Repository	Python ETL/geospatial/scoring pipeline, SQL views, processed/reference artifacts, validation outputs, tests, documentation and Power BI source.
Raw inputs	Not redistributed; obtain them from the original Statistics Canada and Government of British Columbia public sources.

Full project repository: <https://github.com/CarlosNakagomi/Vancouver-Community-Health-Resource-Allocation>